

UNIT 3

Computer Systems (Organization and Architecture)

3.0 Learning Outcomes

After completing this module, you will be able to:

- Solve addition, subtraction, multiplication and division problems in decimal, binary, octal and hexadecimal number system.
- Convert from one radix to another (Decimal, binary, octal and hexadecimal)

3.1 Introduction

We have learned the concept of data storage operations applied to data and output as a results. All these activities (storing, calculating and outputting) are possible because there is a unit assigned to handle each activity. These units are referred to as hardware and the operating system. Computers include the actual computer and the computer peripherals. Computer peripherals are devices that work with the computer.

This unit discussed the computer systems organization and architecture, its components and how it works. In addition, from this unit you will learn the different number systems and how these numbers are converted from one number system to another. Further, the different coding schemes for numeric and non – numeric data will be discussed too.

3.2 Major Components of a Computer System

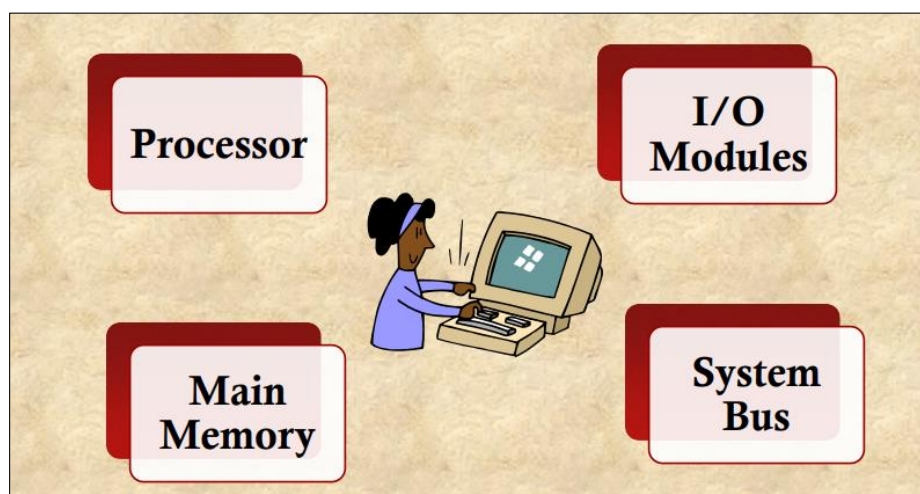


Figure 1 Components of a Computer System

How the System Units Work Together?

When the system units are working together, the results of the computation are periodically being saved in the memory for read – out anytime. The data from outside of the system may be communicated to the system by the input unit. On the other hand, results or data in the system maybe known or stored in another material by the output unit. The users of the machine get to know problems occurring during processing by the output unit too.

The central processing unit controls everything that happens. It is the nerve center of the system. It tells the input when to bring data into the system, or when the output unit would write the information. The CPU decides on when to require input and when to output information based on the set of instructions (called program) in the memory.

Inside the computer, the instructions and information are not recognizable by the human facets because they are all converted to electrical signals of digital codes. These electrical signals are relayed to the different units through the busses. The three principal busses of a computer are: address bus, control bus, and data bus.

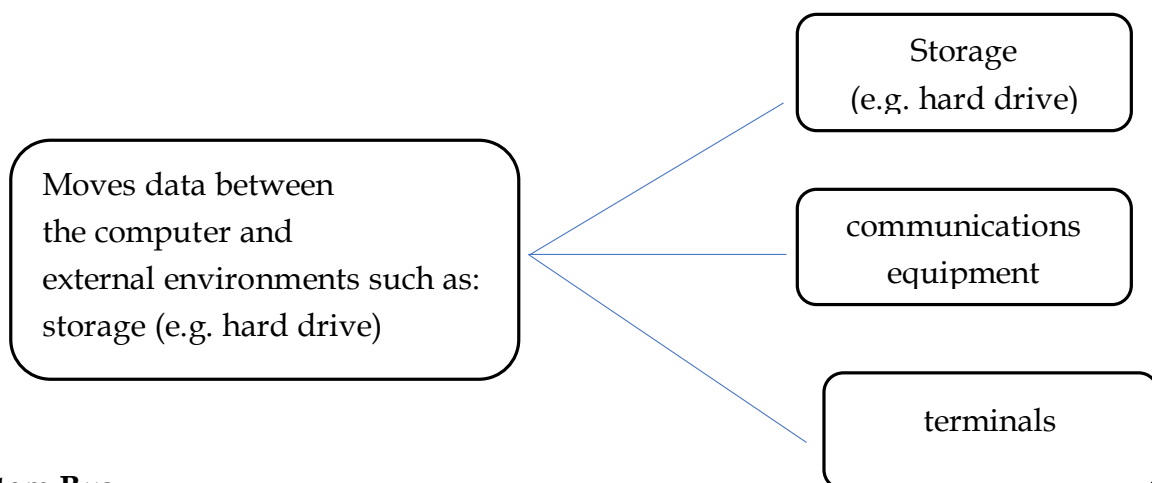
Processor

- Controls the operation of the computer
- Performs the data processing functions
- Referred to as the Central Processing Unit (CPU)

Main Memory

- Volatile
- Contents of the memory is lost when the computer is shut down
- Referred to as real memory or primary memory

I/O Modules



System Bus

- Provides for communication among processors, main memory, and I/O modules

3.2.1 Number System Conversion

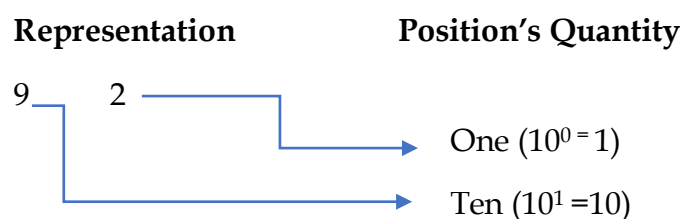
In the number system, each number is represented by its base. If the base is 2 it is a binary number, if the base is 8 it is an octal number, if the base is 10, then it is called decimal number system and if the base is 16, it is part of the hexadecimal number system. The conversion of decimal numbers to any other number system is an easy method. But to convert other base number systems into decimal numbers requires practice.

3.2.2 Decimal Number System

In the **decimal number system**, the numbers are represented with base 10. The way of denoting the decimal numbers with base 10 is also termed as decimal notation. This number system is widely used in computer applications. It is also called the base-10 number system which consists of 10 digits, such as, 0,1,2,3,4,5,6,7,8,9. Each digit in the decimal system has a position and every digit is ten times more significant than the previous digit.

Suppose, 25 is a decimal number, then 2 is ten times more than 5. Some examples of decimal numbers are: $(12)_{10}$, $(345)_{10}$, $(119)_{10}$, $(200)_{10}$, $(313.9)_{10}$. A number system which uses digits from 0 to 9 to represent a number with base 10 is the decimal system number. The number is expressed in base-10 where each value is denoted by 0 or first nine positive integers. Each value in this number system has the place value of power 10. It means the digit at the tens place is ten times greater than the digit at the unit place.

The Decimal System (Base 10 System)



Let us see some more examples:

$$(92)_{10} = 9 \times 10^1 + 2 \times 10^0$$

$$(200)_{10} = 2 \times 10^2 + 0 \times 10^1 + 0 \times 10^0$$

The decimal numbers which have digits present on the right side of the decimal (.) denote each digit with decreasing power of 10.

Some examples are:

$$(30.2)_{10} = 3 \times 10^1 + 0 \times 10^0 + 2 \times 10^{-1}$$

$$(212.367)_{10} = 2 \times 10^2 + 1 \times 10^1 + 2 \times 10^0 + 3 \times 10^{-1} + 6 \times 10^{-2} + 7 \times 10^{-3}$$

3.2.3 Binary Number System

According to digital electronics and mathematics, a binary number is defined as a number that is expressed in the binary system or base 2 numeral system. It describes numeric values by two separate symbols; 1 (one) and 0 (zero). The base-2 system is the positional notation with 2 as a radix.

The binary system is applied internally by almost all latest computers and computer-based devices because of its direct implementation in electronic circuits using logic gates. Every digit is referred to as a **bit**.

What is Bit in Binary Number?

A single binary digit is called a “**Bit**”. A binary number consists of several bits. Examples are:

- 10101 is a five-bit binary number
- 101 is a three-bit binary number
- 100001 is a six-bit binary number

Facts to Remember:

- Binary numbers are made up of only 0's and 1's.
- A binary number is represented with a base-2
- A bit is a single binary digit.

Binary Numbers Table

Some of the binary notations of lists of decimal numbers from 1 to 30, are mentioned in the below list.

<i>Number</i>	<i>Binary Number</i>	<i>Number</i>	<i>Binary Number</i>	<i>Number</i>	<i>Binary Number</i>
1	1	11	1011	21	10101
2	10	12	1100	22	10110
3	11	13	1101	23	10111
4	100	14	1110	24	11000
5	101	15	1111	25	11001
6	110	16	10000	26	11010
7	111	17	10001	27	11011
8	1000	18	10010	28	11100
9	1001	19	10011	29	11101
10	1010	20	10100	30	11110

How to Calculate Binary Numbers

For example, the number to be operated is 1235.

<i>Thousands</i>	<i>Hundreds</i>	<i>Tens</i>	<i>Ones</i>
1	2	3	5

This indicates,

$$1235 = 1 \times 1000 + 2 \times 100 + 3 \times 10 + 5 \times 1$$

Given,

1000	$= 10^3 = 10 \times 10 \times 10$
100	$= 10^2 = 10 \times 10$
10	$= 10^1 = 10$
1	$= 10^0$ (any value to the exponent zero is one)

The above table can be described as,

<i>Thousands</i>	<i>Hundreds</i>	<i>Tens</i>	<i>Ones</i>
10^3	10^2	10^1	10^0
1	2	3	5

Hence,

$$1235 = 1 \times 1000 + 2 \times 100 + 3 \times 10 + 5 \times 1$$

$$= 1 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 5 \times 10^0$$

The decimal number system operates in base 10, wherein the digits 0-9 represent numbers. In binary system operates in base 2 and the digits 0-1 represent numbers, and the base is known as **radix**. Put differently, and the above table can also be shown in the following manner.

<i>Thousands</i>	<i>Hundreds</i>	<i>Tens</i>	<i>Ones</i>	
Decimal	10^3	10^2	10^1	10^0
Binary	2^3	2^2	2^1	2^0

We place the digits in columns 10^0 , 10^1 and so on in base 10. When there is a need to put a value higher than 9 in the form of $10^{(n+1)}$ for instance, to add 10 to column 10^0 , you need to add 1 to the column 10^1 .

We place the digits in columns 2^0 , 2^1 and so on in base 2. To place a value that is higher than 1 in 2^n , you need to add $2^{(n+1)}$. For instance, to add 3 to column 2^0 , you need to add 1 to column 2^1 .

Position in Binary Number System

In the Binary system, we have ones, twos, fours etc...

For example, 1011.110

It is shown like this:

$$1 \times 8 + 0 \times 4 + 1 \times 2 + 1 + 1 \times \frac{1}{2} + 1 \times \frac{1}{4} + 0 \times \frac{1}{8} = 11.75 \text{ in Decimal}$$

To show the values greater than or less than one, the numbers can be placed to the left or right of the point.

For 10.1, 10 is a whole number on the left side of the decimal, and as we move more left, the number place gets bigger (Twice).

The first digit on the right is always Halves $\frac{1}{2}$ and as we move more right, the number gets smaller (half as big).

In the example given above:

- “10” shows ‘2’ in decimal.
- “.1” shows ‘half’.
- So, “10.1” in binary is 2.5 in decimal.

3.2.4 Binary Arithmetic Operations

Like we perform the arithmetic operations in numerals, in the same way, we can perform addition, subtraction, multiplication and division operations on Binary numbers. Let us learn them one by one.

Binary Addition

Adding two binary numbers will give us a binary number itself. It is the simplest method. Addition of two single-digit binary number is given in the table below.

<i>Binary Numbers</i>		<i>Addition</i>
0	0	0
0	1	1
1	0	1
1	1	0; Carry $\rightarrow$ 1

Let us take an example of two binary numbers and add them.

Example: Add 1101_2 and 1001_2 .

Solution:

$$\begin{array}{r} 1101 \\ +1001 \\ \hline 10110 \end{array}$$

Answer: 10110

Binary Subtraction

Subtracting two binary numbers will give us a binary number itself. It is also a straightforward method. Subtraction of two single-digit binary number is given in the table below.

<i>Binary Numbers</i>		<i>Subtraction</i>
0	0	0
0	1	1; Borrow 1
1	0	1
1	1	0

Let us take an example of two binary numbers and subtract them.

Example: Subtract 1101_2 and 1010_2 .

Solution:

$$\begin{array}{r} 1101 \\ - 1010 \end{array}$$

Answer: 0011

Binary Multiplication

The multiplication process is the same for the binary numbers as it is for numerals. Let us understand it with example.

Example: Multiply 1101_2 and 1010_2 .

Solution:

$$\begin{array}{r}
 1101 \\
 \times 1010 \\
 \hline
 0000 \\
 1101 \\
 0000 \\
 1101 \\
 \hline
 10000010
 \end{array}$$

Answer: 10000010

Binary Division

The binary division is similar to the decimal number division method. We will learn with an example here.

Example: Divide 1010_2 by 10_2

Solution:

$$\begin{array}{r}
 10 \overline{)1010} (101 \\
 \underline{10} \\
 010 \\
 \underline{10} \\
 0
 \end{array}$$

Answer: 0

Uses of Binary Number System

Binary numbers are commonly used in computer applications. All the coding and languages in computers such as C, C++, Java, etc. use binary digits 0 and 1 to write a program or encode any digital data. The computer understands only the coded language. Therefore, these 2-digit number system is used to represent a set of data or

information in discrete bits of information.

Problems and Solutions

Let us practice some of the problems for better understanding:

Question 1: What is binary number 1.1 in decimal?

Solution:

Step 1: 1 on the left-hand side is on the one's position, so it's 1.

Step 2: The one on the right-hand side is in halves, so it's

$$1 \times \frac{1}{2}$$

Step 3: so, $1.1 = 1.5$ in decimal.

Question 2: Write 10.11_2 in Decimal?

Solution:

$$10.11 = 1 \times (2)^1 + 0 (2)^0 + 1 (\frac{1}{2})^1 + 1(\frac{1}{2})^2$$

$$= 2 + 0 + \frac{1}{2} + \frac{1}{2}$$

$$= 2.75$$

So, 10.11 is 2.75 in Decimal.

How to convert a decimal number into a binary number?

To convert a decimal number into its equivalent binary number, we divide the decimal number by 2 each time, till we get 0 as a dividend. Let us take an example to convert 1310 into a binary number.

$$13 \div 2: 6 \text{ and remainder } 1$$

$$6 \div 2: 3 \text{ and remainder } 0$$

$$3 \div 2: 1 \text{ and remainder } 1$$

$$1 \div 2: 0 \text{ and remainder } 1$$

Now we take the bits from the last remainder to first remainder, i.e.(MSB to LSB). Hence, $13_{10} = 1101_2$

What is the use of binary numbers?

Binary numbers are commonly used in computer architecture. Since the computer understands only the language of two digits 0's and 1's, therefore the programming is done using a binary number system.

3.2.5 Octal Number System

A number system which has its base as 'eight' is called an Octal number system. It uses numbers from 0 to 7. Let us take an example, to understand the concept. As we said, any number with base 8 is an octal number like 24_8 , 109_8 , 55_8 , etc.

Like Octal number is represented with base 8, in the same way, a binary number is represented with base 2, decimal number with base 10 and the hexadecimal number is represented with base 16. *Examples for these number systems are:*

- 22_2 is a binary number
- 100_{10} is a decimal number
- 40_{16} is a hexadecimal number

If we solve an octal number, each place is a power of eight.

- $124_8 = 1 \times 8^2 + 2 \times 8^1 + 4 \times 8^0$

Octal Numbers Chart

We use only 3 bits to represent Octal Numbers. Each group will have a distinct value between 000 and 111.

<i>Octal Digital Value</i>	<i>Binary Equivalent</i>
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Decimal to Octal Number

To convert decimal to octal number, octal dabble method is used. In this method, the decimal number is divided by 8 each time, it yields or gives a remainder. The first remainder we get is the least significant digit(LSD) and the last remainder is the most significant digit(MSD). Let us understand the conversion with the help example.

Solved Example

Problem 1: Suppose 560 is a decimal number. Convert it into an octal number.

Solution: If 560 is a decimal number, then,

$560/8=70$ and remainder is 0

$70/8=8$ and remainder is 6

$8/8=1$ and remainder is 0

And $1/8=0$ and remainder is 1

So the octal number starts from MSD to LSD, i.e. 1060

Therefore, $560_{10} = 1060_8$

Problem 2: Convert 0.52 into an octal number.

Solution: The fraction part of the decimal number has to be multiplied by 8.

$0.52 \times 8=0.16$ with carry 4

$0.16 \times 8=0.28$ with carry 1

$0.28 \times 8=0.24$ with carry 2

$0.24 \times 8=0.92$ with carry 1

So, for the fractional octal number, we read the generated carry from up to down.

Therefore, 4121 is the octal number.

Octal to Decimal Number

Let us learn here, conversion of Octal number to Decimal Number or base 8 to base 10.

Solved Example

Example 1: Suppose 215_8 is an octal number, then it's decimal form will be,

$$\begin{aligned} 215_8 &= 2 \times 8^2 + 1 \times 8^1 + 5 \times 8^0 \\ &= 2 \times 64 + 1 \times 8 + 5 \times 1 = 128 + 8 + 5 \\ &= 141_{10} \end{aligned}$$

Example 1: Let 125 is an octal number denoted by 125_8 . Find the decimal number.

$$\begin{aligned} 125_8 &= 1 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 \\ &= 1 \times 64 + 2 \times 8 + 5 \times 1 = 64 + 16 + 5 \\ &= 85_{10} \end{aligned}$$

Binary to Octal Number

A binary number can be converted into an octal number, with the help of the below-given table.

Octal Number	Equivalent Binary Number
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111

Solved Example

Example: Convert $(100010)_2$ to octal number.

Solution: With the help of the table we can write,

$100 \rightarrow 4$

and $010 \rightarrow 2$

Therefore, $(100010)_2 = 42$

Similarly, we can convert an octal number to binary number with the help of the table.

Octal to Hexadecimal Number

Hexadecimal number consist of numbers and alphabets. It is represented with base 16. The numbers from 0-9 are represented in the usual form, but from 10 to 15, it is denoted as A, B, C, D, E, F. Conversion of the octal number to hexadecimal requires two steps.

- First, convert octal number to decimal number.
- Then, convert decimal number to hexadecimal number.

Let us understood with the help of an example. We will take the same example, where we have converted octal number to decimal, such as;

$$(215)_8 = (45)_{10}$$

Now, convert $(45)_{10}$ into a hexadecimal number by dividing 45 by 16 until you get remainder less than 16.

$$\begin{array}{r|l}
 16 & 45 \\
 \hline
 16 & 2 \text{ ---} 13 \\
 \hline
 & 0 \text{ ---} 2
 \end{array}$$

Therefore, we can write, $(45)_{10} = (2D)_{16}$

Or $(215)_8 = (2D)_{16}$

Octal Multiplication Table

*	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	10	12	14	16
3	0	3	6	11	14	17	22	25
4	0	4	10	14	20	24	30	34
5	0	5	12	17	24	31	36	43
6	0	6	14	22	30	36	44	52
7	0	7	16	25	34	43	52	61

Applications

The octal Number system is widely used in computer application sectors and also in the aviation sector to use the number in the form of code.

Based on octal number system applications, several computing systems are developed. All the modern generation computing system uses 16-bit, 32-bit or 64-bit word which is further divided into 8-bit words. Similarly, for various programming languages, octal numbers are used to do coding or to write the encrypted language, which is only understood by the computing machine.

Also in the aviation sector or field or say aviation industry, Transponders used in the aircraft transmits a code which is expressed as four octal digit number. These codes are interrogated by ground radar.

3.2.6 Hexadecimal Number System Conversions

As we know, there are 16 digits in hexadecimal numbers, represented from 0 to 9 same like decimals, but after that, it starts with an alphabetical representation of preceding numbers such as A, B, C, D and E. Let us see the conversion of 'hex' into other number systems.

Hexadecimal to Decimal Number System Conversion

Here, you will see the representation of a hexadecimal number into decimal form.

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Decimal to Hexadecimal Number System Conversion

You have learned how to convert hexadecimal number to decimal number. Now let us find out how we can convert a decimal number in hexadecimal.

Follow the below steps:

- Firstly, divide the number by 16
- Take the quotient and divide again by 16
- The remainder left will produce the hex value
- Repeats the steps until the quotient has become 0

Example: Convert $(242)_{10}$ into hexadecimal.

Solution: Divide 242 by 16 and repeat the steps, till the quotient is left as 0.

$$\begin{array}{r|l}
 16 & 242 \\
 \hline
 16 & 15 \quad 2 \rightarrow 2 \\
 \hline
 & 0 \quad 15 \rightarrow F
 \end{array}$$

Therefore, $(242)_{10} = (F2)_{16}$

Hexadecimal to Octal Number System Conversion

Here, you will see the representation of a hexadecimal number into octal number form.

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17

Octal to Hexadecimal Number System Conversion

To convert octal to hex, we have to first convert octal number to decimal and then decimal to hexadecimal. Let us understand it with the help of an example;

Example: Convert $(121)_8$ into hexadecimal.

Solution: First convert 121 into decimal number.

$$\Rightarrow 1 \times 8^2 + 2 \times 8^1 + 1 \times 8^0$$

$$\Rightarrow 1 \times 64 + 2 \times 8 + 1 \times 1$$

$$\Rightarrow 64 + 16 + 1$$

$$\Rightarrow 81$$

$$(121)_8 = 81_{10}$$

Now converting 81_{10} into a hexadecimal number.

$$\begin{array}{r|l} 16 & 81 \\ \hline 16 & 5 \quad 1 \rightarrow 1 \\ \hline & 0 \quad 5 \rightarrow 5 \end{array}$$

Therefore, $81_{10} = 51_{16}$

Hexadecimal to Binary Number System Conversion

Here, you will see the representation of a hexadecimal number into binary form. We can use only 4 digits to represent each hexadecimal number, where each group has a distinct value from 0000 (for 0) and 1111 (for F= 15 =8 + 4 + 2 + 1).

Hexa-decimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Binary to Hexadecimal Number System Conversion

Binary to hexadecimal conversion is a simple method to do. You just have to put the values of the binary number to the relevant hexadecimal number.

Example: Convert $(11100011)_2$ to hexadecimal.

Solution: From the table, we can write, 11100011 as E3.

Therefore, $(11100011)_2 = (E3)_{16}$

Hexadecimal Number System Conversion Examples

Example 1: What is 5C6 (Hexadecimal)?

Solution: Step 1: The "5" is the "16 x 16" position, so that means $5 \times 16 \times 16$

Step 2: The 'C' (12) is in the "16" position, so that means 12×16 .

Step 3: The "6" in the "1" position so that means 6.

Answer is : $5C6 = 5 \times 16 \times 16 + 12 \times 16 + 6 = (1478)$ in Decimal.

Example 2: What is 3C5 (Hexadecimal)?

Solution: Step 1: The "3" is the "16 x 16" position, so that means $3 \times 16 \times 16$

Step 2: The 'C' (12) is in the "16" position, so that means 12×16 .

Step 3: The "5" in the "1" position so that means 5.

Answer is : $3C5 = 3 \times 16 \times 16 + 12 \times 16 + 5 = (965)$ in Decimal.

Example 3: What is 7B5 (Hexadecimal)?

Solution: Step 1: The "7" is the "16 x 16" position, so that means $7 \times 16 \times 16$

Step 2: The 'B' (11) is in the "16" position, so that means 11×16 .

Step 3: The "5" in the "1" position so that means 5.

Answer is : $7B5 = 7 \times 16 \times 16 + 11 \times 16 + 5 = (1973)$ in Decimal.

Example 4: What is 2E8 (Hexadecimal)?

Solution: Step 1: The "2" is the "16 x 16" position, so that means $2 \times 16 \times 16$

Step 2: The 'E' (14) is in the "16" position, so that means 14×16 .

Step 3: The "8" in the "1" position so that means 8.

Answer is : $2E8 = 2 \times 16 \times 16 + 14 \times 16 + 8 = (744)$ in Decimal.

Example 5: What is 4F8 (Hexadecimal)?

Solution: Step 1: The "4" is the "16 x 16" position, so that means $4 \times 16 \times 16$

Step 2: The 'F' (15) is in the "16" position, so that means 15×16 .

Step 3: The "8" in the "1" position so that means 8.

Answer is : $4F8 = 4 \times 16 \times 16 + 15 \times 16 + 8 = (1272)$ in Decimal.

Test your Knowledge:

What is 5D6 (Hexadecimal)?

Notes:

- Out of many types of number representation techniques, Hexadecimal number system is one having a value of base 16.
- So Hexadecimal numbers have 16 symbols or digital values, i.e 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
- A, B, C, D, E, F are single bit representations of 10, 11, 12, 13, 14 and 15 respectively.
- Addition of either an *o* prefix or an *h* prefix indicates Hexadecimal. A power of 16 is the weight of the position of every digit.



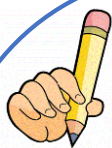
For further understanding with the topics in this unit, please visit
https://www.youtube.com/watch?v=L2zsmYal5ww&ab_channel=TheOrganicChemistryTutor

3.3 References

https://www.youtube.com/watch?v=L2zsmYal5ww&ab_channel=TheOrganicChemistryTutor
<https://byjus.com/maths/decimal-number-system/>
<https://byjus.com/maths/binary-number-system/>
<https://byjus.com/maths/octal-number-system/>
<https://byjus.com/maths/hexadecimal-number-system/>
<https://www.unf.edu/public/cop4610/ree/Notes/PPT/PPT8E/CH%2001%20-OS8e.pdf>

3.4 Acknowledgement

All the figures and information presented in this module were taken from the references enumerated above.

**ASSESSMENT**

Name: _____ Year, Course and Section: _____

Subject: _____ Instructor: _____

I. Direction: Please answer the following questions. Be reminded that you need to show your solutions in every answered item. Score will not be counted if you fail to show/attached your solutions.

1. Convert each of the following binary representations to its equivalent decimal forms.
 - b. 0101
 - c. 0110
 - d. 10010
 - e. 111011
 - f. 1101101
2. Convert each of the following decimal representations to its equivalent forms.
 - a. 13
 - b. 27
 - c. 5
 - d. 156
 - e. 485
3. Add the following binary representation.
 - a. $110101 + 1101101$
 - b. $1101 + 1010$
 - c. $100111 + 111100$
4. Subtract the following binary representation.
 - a. $1101101 - 110101$
 - b. $1101 - 1010$
 - c. $111100 - 100111$
5. Multiply the following binary representation.
 - a. 110101×1101101
 - b. 1101×1010
 - c. 100111×111100

6. Divide the following binary representation.
- 1101101/110101
 - 1101/1010
7. Convert the following decimal representation to binary, octal and hexadecimal form.

Decimal	Binary	Octal	Hexadecimal
65			
38			
89			
46			
51			

II. Create a video presentation wherein you demonstrating on how to convert decimal - binary - octal - hexadecimal and vice - versa. Please refer from item #7 for the numbers that needs to be converted.